MICHAEL SHAO

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PROFESSIONAL SUMMARY

Senior Software Engineer at Playlist Technologies with 10+ years of distributed systems experience at Amazon and AWS. Previously led legal technology infrastructure and agentic AI workflows at Morgan & Morgan. Expertise in Python/FastAPI, cloud infrastructure, Datadog observability, and Terraform/Terragrunt. AWS Databases Blog author and re:Invent 2024 speaker, with experience setting technical standards, interviewing, and mentoring engineers.

CORE COMPETENCIES

Technical Leadership: Architectural Design, System Design Reviews, Technical Mentoring, Code Quality Standards, 0-1 Experimentation

Backend Systems: Distributed Systems, Microservices, RESTful APIs, Asynchronous Processing, Event-Driven Architecture

Cloud & Infrastructure: AWS (DynamoDB, Lambda, API Gateway, SNS/SQS, SWF, CDK), Terraform, Terragrunt, Infrastructure as Code

AI / LLM Tooling: LLM Orchestration, Agentic Workflows, Claude/Anthropic, ML-Powered Data Processing

Observability: Datadog (APM, Dashboards, Alerting), Structured Logging

Data Storage: DynamoDB, DocumentDB, Elastic Cluster Operations, Document Indexing, Dyntastic, Data Pipeline Architecture

Languages: Python, FastAPI, Pydantic, Java, JavaScript/TypeScript

Cross-Functional: Developer Advocacy, Technical Content Strategy, Product Management Advisory, Technical Bar Raiser

PROFESSIONAL EXPERIENCE

Playlist Technologies - Senior Software Engineer

Aug 25, 2026 - Present

Morgan & Morgan, P.A. - Staff Software Engineer / Tech Lead Manager (TLM)

Remote (Redmond, WA) - Jul 2025 - Aug 21, 2026

Led 0-1 experimentation initiatives and human-in-the-loop productivity improvements for a legal technology platform serving Mass Tort, Medical Malpractice, and Product Liability cases. Direct leadership of a 2-person team with indirect influence across a 30-person engineering organization.

- **Agentic AI platform:** Architected LLM-powered agentic workflows using Claude Code (Anthropic) and Python/FastAPI, establishing coding standards and agentic patterns adopted across a 30-person engineering org - measurably reduced review cycles and accelerated feature delivery
- **Document indexing and search:** Owned document indexing services and operations for Elastic clusters holding medical records, court dockets, and police reports, supporting document search through the firm's customer-facing AI tooling and its internal model facade
- **Client data processing:** Designed and implemented ML-powered address parsing system matching 2,250 clients to contaminated water sites with 89% accuracy for year-range address mappings, eliminating a manual data-entry bottleneck
- **Observability:** Instrumented backend services with Datadog APM and structured logging, establishing team-wide dashboards and alerting standards that introduced proactive incident detection where none previously existed
- **Infrastructure as code:** Owned Terraform/Terragrunt configuration across 7 services (including 2 greenfield builds), enabling repeatable environment provisioning and auditability across the platform
- **Build performance:** Cut primary repository test cycle time by 50% through remote caching, accelerating developer iteration
- **CI/CD reliability:** Improved deployment reliability and rollback safety by migrating to version-controlled release tagging, reducing mean time to recovery

PROFESSIONAL EXPERIENCE (CONTINUED)

Amazon Web Services (AWS) - Senior Developer Advocate, DynamoDB

Seattle, WA - Nov 2023 - Jul 2025

(Technical Interviewing Bar Raiser: 200+ interviews; Asians@Amazon Global Board, Professional Development Coordinator; Social Media Bar Raiser) Led technical content and developer experience initiatives for Amazon DynamoDB while maintaining engineering responsibilities. Bridged product engineering, marketing, and the developer community through strategic technical storytelling.

- **Automated publishing pipeline:** Architected automated code sample integration system for the AWS SDK Examples repository, reducing manual operations by 75% and improving code quality consistency across the DynamoDB developer experience
- **Cross-team standardization:** Led internal engineering effort to standardize documentation and code sample creation processes across 5 teams, enabling faster iteration
- **Developer reach:** Published technical deep-dives on CDC strategies and DynamoDB 2024 innovations on the AWS Databases Blog, driving measurable developer awareness
- **re:**Invent 2024 speaker: Delivered session DAT414 (DynamoDB Code Talk) at AWS re:Invent 2024 to 50+ engineers
- **Strategic advisory:** Served as technical advisor to the non-relational databases Product Management team, influencing roadmap decisions based on customer feedback

Amazon.com - Senior Software Engineer, Retail Compatibility Experience (ARCX)

Seattle, WA - Jul 2022 - Nov 2023

Led engineering efforts for a compatibility service platform processing millions of customer queries daily. Spearheaded technical design and launch execution for critical retail infrastructure during the Unified Compatibility (Unicom) initiative.

- **ARCAPI:** Architected the ARCAPI service backend from 0-1, handling millions of daily retail compatibility queries at sub-100ms p99 latency across 1M+ compatibility records, driving the Unicom initiative from design through production launch
- **Technical leadership:** Grew team throughput by 25% by owning system design review cycles and running structured mentorship for entry-level engineers, with 4 promoting to the next level

Amazon.com - Software Development Engineer, Confident Customer Shopping (CCS)

Seattle, WA - Dec 2019 - Jul 2022

Architected data pipelines and front-end widgets surfacing compatibility information. Executed on a mission to accelerate feature development through scalable platform solutions.

- **Database migration:** Led zero-downtime migration of critical compatibility data from DocumentDB to DynamoDB, eliminating 90% of operational overhead
- **API performance:** Designed Lambda-based RESTful service improving API throughput by 40% and absorbing peak traffic load without degradation

Amazon.com - Software Development Engineer, Supplier Fulfillment Services (SFS)

Seattle, WA - Jun 2015 - Dec 2019

Designed and implemented critical backend services for a third-party seller logistics platform connecting Seller Central merchants with DHL, DPD, FedEx, UPS, USPS, and Royal Mail, managing millions of daily transactions.

- **Reliability engineering:** Designed self-throttling transaction persistence layer (DynamoDB + AWS SWF) sustaining burst load while protecting downstream carrier integrations (DHL, FedEx, UPS) from burst-load overload
- **Async dispatch:** Designed configuration-driven manifest dispatcher using AWS SNS/SQS, reducing latent operational calls by 30%

EDUCATION

University of Waterloo, Faculty of Mathematics - Honours Bachelor of Computer Science (Co-op Program), Minor in History - 2014